

Objective :

The objective of this experiment is to implement some circuit diagrams with basic gates.

problem specification:

We have to implement the following functions with basic gates.

a) $F(P, Q, R) = PR + \bar{Q}R$

b) $F(P, Q, R) = PR(PQ + QR)$

c) $F(P, Q, R) = \bar{P} + Q + R$

Required instruments:

No	Name	Quantity
1.	Trainer Board	1 piece
2.	IC - Extractor	1 piece
3.	IC (Hex-Inverter) (7404)	1 piece
4.	IC (Quad 2 input AND) (7408)	1 piece
5.	IC (Quad 2 input OR) (7432)	1 piece
6.	Hand Book	1 piece
7.	Wires	A lot

Truth Table :-

a) $F(P, Q, R) = PR + \bar{Q}R$

P	Q	R	$\bar{Q}$	PR	$\bar{Q}R$	$PR + \bar{Q}R$
0	0	0	1	0	0	0
0	0	1	1	0	1	1
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	0	1	1	1	1	1
1	1	0	0	0	0	0
1	1	1	0	1	0	1

P.T.O →

$$\begin{aligned} b) F(P, Q, R) &= PR (PQ + QR) \\ &= PPQR + PRQR \\ &= PQR + PQR \\ &= PQR \end{aligned}$$

P	Q	R	PQR
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

P.T.O →

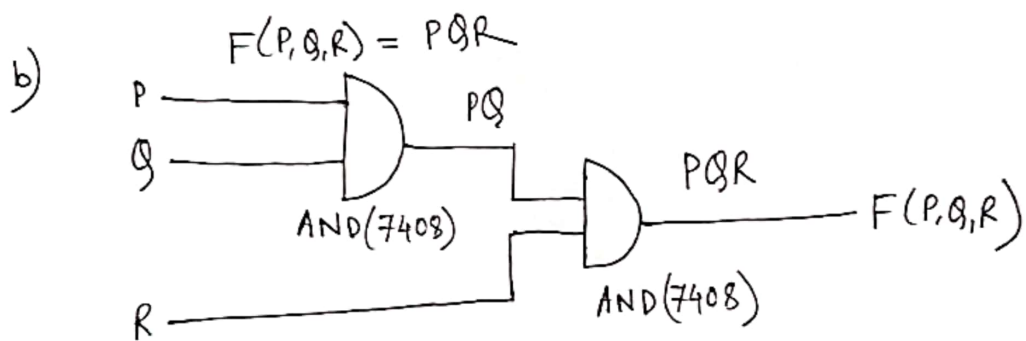
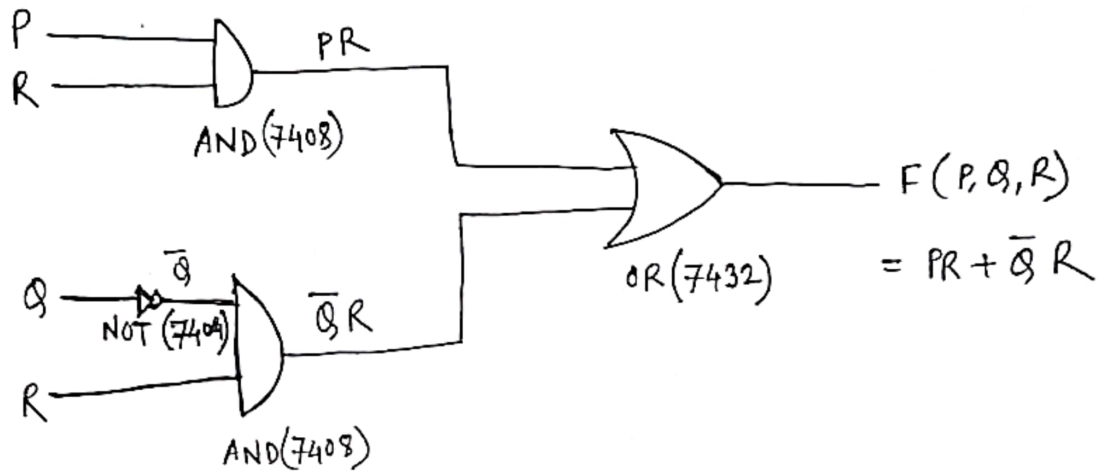
c) $F(P, Q, R) = \bar{P} + Q + R$

P	Q	R	$\bar{P}$	$\bar{P} + Q + R$
0	0	0	1	1
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

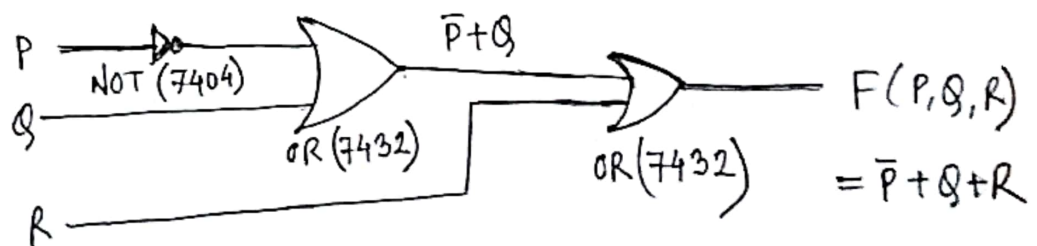
P.T.O.

Circuit Diagram

a) $F(P, Q, R) = PR + \bar{Q}R$



c) $F(P, Q, R) = \bar{P} + Q + R$



Discussion

All the diagrams were drawn from the simplified forms of given functions and these were verified by the teachers. Truth table was also verified according to experiment. During the experiment, all the circuits were implemented according to the diagrams. All the ICs were tested before the final implementations. The wires used for connecting the ICs were used in such a way that ensured the implementation not to be clumsy. The experiment was verified by the output of the truth table. No major problem occurred during the experiment.